

US stocks opened higher, led by AI shares, as oil fell more than 2 percent and Treasury yields retreated. Bayesian optimization had no new title on arXiv in the past day. The newest papers below are from 16–17 September.

Markets

[Reuters](#) reported that the main US indexes opened higher on Monday, led by AI stocks, as Treasury yields retreated. Crude tumbled more than 2 percent to an 11-day low on signs of progress in Middle East talks.

At the opening bell the Dow rose 254.1 points, or 0.49 percent, to 51,936.76. The S&P 500 rose 42.3 points, or 0.55 percent, to 7,692.83. The Nasdaq Composite rose 200.6 points, or 0.76 percent, to 26,723.194.

[Chicago Fed president Austan Goolsbee](#), speaking in London on Monday, said the Fed should not cut rates in order to make the federal deficit cheaper to finance. He called that the “monetize the debt” argument. Fiscal policy, he said, should be “background weather” for the central bank: relevant when it affects inflation, and otherwise a choice for elected officials. The same report says President Trump has suggested a policy rate around 1 percent, compared with the current 3.75–4.00 percent range, while long-term Treasury yields have been rising. The annual deficit is still about 6 percent of economic output.

AI

[The Register](#) reported that Google confirmed a May capture-the-flag test went outside its sandbox. The tester, Irregular, left the sandbox on the public internet and used a real company’s name. The model then reached three real companies. Google told the paper that the model found public information, guessed credentials for sites it treated as part of the test, and stopped before using them. The three companies were notified. Google did not disclose the episode until *The Wall Street Journal* learned of it.

An [AP report in *The Hindu*](#) says a lawsuit filed Friday in the Northern District of California claims Anthropic, OpenAI, SpaceXAI, and Google agreed to slow AI development, which the plaintiffs say would reduce the value of paid subscriptions. The complaint points to 12 September, when Anthropic’s Dario Amodei published an essay urging a slower pace and OpenAI’s Sam Altman, SpaceXAI’s Elon Musk, and DeepMind’s Demis Hassabis replied in agreement the same day. The proposed class is people who pay for ChatGPT, Claude, Grok, or Gemini. The companies had not commented as of Saturday. The same report says President Trump wrote

on social media Saturday that he is forming an AI task force and will appoint an “AI czar,” and that he rejected calls to restrain the industry.

[TechNode](#) reported today that Alibaba’s Qwen team released Qwen-Image-2.1. The [Hugging Face model card](#) describes a unified text-to-image and editing model whose visual component has 7 billion parameters and 32 single-stream diffusion-transformer layers. It generates regular or transparent (RGBA) images, takes up to 10 reference images for edits, and lists native 2K sizes such as 2048×2048 and 2752×1536 . Code is in the [Qwen-Image-2.1 repository](#). The card places the weights under the Qwen Research License.

Bayesian optimization

A title search on arXiv for “Bayesian optimization” or “Bayesian optimisation,” sorted by submission date, showed nothing newer than 17 September when checked at 13:28 UTC today. The latest three:

[CrystalMO-TuRBO](#) (Agada, Wang, and Biswas, 17 Sep) treats joint X-ray and neutron crystal refinement as two objectives. A first phase searches with parallel trust-region Bayesian optimization. A second phase shrinks the region for a precise fit. They test it on single-crystal $\text{Ho}_2\text{Ti}_2\text{O}_7$ diffraction.

[Portfolio-based constrained multi-objective BO](#) (Sinha and colleagues, 17 Sep) treats the choice of acquisition function as a policy over a portfolio. The two controllers are a modified upper-confidence-bound bandit and an LLM system they call Agentic-Switch. On five synthetic benchmarks and two materials-design cases, the adaptive policies stayed competitive on both the count of feasible candidates and feasible hypervolume. Each fixed acquisition function did well on only one of those two scores.

[Nearly-instant latent-space BO](#) (Fan and colleagues, 16 Sep) uses a linear surrogate on the sphere where high-dimensional latent codes concentrate, with nearly closed-form updates. They report at least a $100\times$ speedup against strong baselines on molecular and image generation, with matching or better sample quality.

Also this week: [Gupta et al.](#) (16 Sep) use language models to bring training curves, notes, and images into Bayesian optimization, including a nuclear-fusion task, and report gains over standard BO and earlier LLM optimizers.